

HomeNet Sentinel

Lightweight SIEM + Wireless Threat Detection System

Executive Summary

HomeNet Sentinel is a distributed wireless intrusion detection system (WIDS) designed to bridge IoT hardware and centralized security analytics. By utilizing ESP32 nodes as remote sensors and a Python-based backend for log ingestion and heuristic analysis, the system identifies rogue access points, security downgrades, and anomalous network activity within a home or small office environment.

System Architecture

- **Sensor Layer:** ESP32 microcontrollers performing continuous 802.11 beacon frame scanning.
- **Ingestion Layer:** FastAPI (Python) asynchronous REST API hosted on TrueNAS.
- **Intelligence Layer:** Heuristic Rule Engine for "Evil Twin" and security policy violation detection.
- **Storage Layer:** SQLite (MVP) for historical network mapping and alert persistence.
- **Presentation Layer:** Web Dashboard for real-time visualization of the wireless landscape.

Initial Development Roadmap (Sprints 1-4)

Phase	Objective	Key Deliverables
Sprint 1: Foundation	API & Schema Setup	FastAPI Skeleton, Pydantic Data Validation, POST /scan endpoint.
Sprint 2: Persistence	Database Logic	SQLAlchemy Integration, Scans and KnownDevices tables.
Sprint 3: SIEM Brain	Detection Engine	Evil Twin logic, Encryption downgrade alerts, Security policy enforcement.
Sprint 4: Visibility	Simulation & Query	GET /alerts, Historical query logic, Python-based Mock Sender script.

Technical Specification

Primary Stack: Python 3.10+, FastAPI, SQLAlchemy, Uvicorn.

Target Environment: TrueNAS (Docker Container), ESP32 (Arduino Framework).

License: Apache 2.0 (To ensure patent protection and permissive use).

Repository Strategy: Monorepo with /backend, /firmware, and /dashboard directories.

Future Evolution

Phase 5: Hardware Integration

Deploying physical ESP32 sensors to transmit real-world 2.4GHz environment data via HTTP JSON payloads.

Phase 4: Advanced Detection (The SIEM Expansion)

Integrating non-wireless logs (SSH attempts, Auth logs) from the TrueNAS host to correlate physical proximity with digital intrusion attempts.

Phase 7: Real-time Alerting

Webhooks for Discord, Slack, or Email notifications when high-severity threats (Evil Twins) are identified.

| Security & Compliance

The project will maintain a strict `.gitignore` to prevent the leakage of local network data (SSIDs/MACs) or TrueNAS credentials. All sensitive configuration will be managed via `.env` files.